

## FastAPI CRUD Q1 Answers

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### 1. What is CRUD in FastAPI ?

CRUD in FastAPI refers to the basic operations performed on data.

- C – Create will add new data using POST method
  - R – Read will fetch data using GET method.
  - U – Update will modify the existing data using PUT method.
  - D – Delete will remove data using DELETE method.
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### 2. Why do we use the POST method for inserting data ?

We use the POST method to insert data because it sends new data from the client to the server. Post is mainly designed for creating new records in databases. In FastAPI, data entered by the user is sent in the request body using POST. Example, Adding a new student, user into the system.

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### 3. Difference between PUT and PATCH

- PUT is used to update the entire record. Usually send all fields again. And it replaces the old data completely.
  - PATCH is used to update only specific fields of a record/ It modifies only the required part.
  - Example, PUT updates full student details but PATCH updates only marks or name.
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### 4. What is the path parameter ?

A path parameter is a value passed inside the URL to identify a specific resource. It is used to access, update or delete particular data. In FastAPI, path parameters are defined inside {} in the route. Path parameters make APIs dynamic because different values in the URL can return different results.

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### 5. What is the request body in FastAPI?

The request body in FastAPI is the data sent by the client to the server while making a request. It is mainly used with POST, PUT and PATCH methods to send information like Json data. FastAPI uses pydantic models to validate and process the request body automatically.

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## 6. Why do we use Pydantic models?

Pydantic models are used to define the structure of input and output data in FastAPI. They automatically validate the data types and ensure correct user input. They help convert json data into python objects easily. It also improves code readability and generate automatic API documentation.

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## 7. What status code is returned for successful deletion ?

The status code returned for successful deletion is usually 404 no content. It means the resource was deleted successfully and no response body is returned. Sometimes APIs also use 200 OK when a success message is returned after deletion.

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## 8. How can we check whether a student exists before updating ?

We can check whether a student exists by searching the student ID in the database or list before updating. If the ID is found, update the record, otherwise raise an error like 404 not found.

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## 9. What happens if duplicate IDs are inserted ?

If duplicate IDs are inserted, multiple records may have the same identifier, causing confusion while fetching, updating or deleting data. The system may return incorrect or unexpected results because IDs are supposed to be unique. So, we usually check whether the ID already exists before inserting new data. If a duplicate ID is found, we raise an error like 400 Bad Request.

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## 10. How does FastAPI automatically generate Swagger UI ?

FastAPI automatically generate Swagger UI using the metadata from your API routes, Pydantic models, and type hints. It reads details like endpoints, request bodies, response models, and HTTP methods from the code. FastAPI then creates interactive API documentation based on the OpenAPI standard. The SwaggerUI is automatically available at /docs.

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THE END